

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

NATURAL RUBBER PRODUCTION
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Give an example for a major yield stimulant used in extraction of latex?	M 1.04	R
2	Clones developed from mother tree of unknown parentage are known as..... ?	M 1.02	R
3	List any two processes proposed for concentrating NR latex?	M 2.03	R
4	Name any two centrifuging machines used in latex concentrating industries?	M 2.04	R
5	List different types of sheet rubbers?	M 3.02	R
6	Expand the abbreviation ISNR?	M 3.04	R
7	List different grades of ISNR?	M 3.04	R
8	PA 80 contains..... parts of vulcanised andparts of unvulcanised rubber?	M 4.02	R
9	Expand the abbreviation of OENR?	M 4.04	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	List at least three major sources of NR with it's popular names?	M1.01	R
2	What are the steps of processing skim latex into skim rubber?	M2.04	U
3	Give a short note on the manufacturing process of Air Dried Sheets?	M3.02	U

4	Write down the main steps in the manufacturing process of RSS?	M3.02	U
5	List the parameters of No.1X RSS as per the Green Book?	M3.02	R
6	What do you mean by PLC and write down different grades of it?	M3.03	R
7	Write any one method of preparing OENR?	M4.04	U
8	Write down the combination of Tyre rubber?	M4.04	R
9	Give the short description of Reclamator Process for manufacturing Reclaimed Rubber?	M4.05	U
10	How do you prepare SP rubber?	M4.02	U

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Outline the methods of application of yield stimulants?	M 1.04	R
OR			
2	Describe the different types of tapping and it's significance?	M 1.03	U
OR			
3	What are the steps of applying rain-guards in rubber trees?	M 1.04	U
OR			
4	Explain different classifications of clones based on the origin?	M 1.02	U
OR			
5	Explain the layout of a centrifuging plant for a medium scale industry?	M2.03	U
OR			
6	Mention the BIS specifications of various quality parameters of centrifuged latex?	M2.03	R
OR			
7	Describe the creaming process for concentration of NR latex?	M2.03	U
OR			

8	Explain the production of double centrifuged latex and it's applications?	M2.04	U
9	Describe the machinery and process of manufacture of ISNR 5?	M3.04	U
OR			
10	Chart the BIS specifications for ISNR grades?	M3.04	R
11	Summarize the production process of CV rubber?	M4.03	U
OR			
12	Explain the preparation and applications of LVRubber?	M4.03	U

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SCHEME OF VALUATION

(SCORING INDICATORS)

Revision : 2021					Course Code :				
Course Title : Natural rubber production									
Qn. No.	SCORING INDICATORS				SPLIT UP SCORE	SUB TOTAL		TOTAL	
I	PART A								
1.	Etheophone				1	1		1	
2.	Primary clones				1	1		1	

3.	Centrifuging, creaming	0.5*2	1	1
4.	Alpha Laval, De Laval	0.5*2	1	1
5.	Ribbed Smoked Sheet, Air Dried Sheet	0.5*2	1	1
6.	Indian Standard Natural Rubber	1	1	1
7.	ISNR 3CV, ISNR 3L, ISNR 5, ISNR 10, ISNR 20, ISNR 50	1	1	1
8	80 and 20	1	1	1
9	Oil Extended Natural rubber	1	1	1
II	PART B			
1.	Species HeveaBrasiliensisPara rubber PartheniumArgentatum Gauyule rubber ManihotGlazioviiCeara Rubber FicusElasticaIndia Rubber CastillaElasticaPanama Rubber (Any three)	1*3	3	3
2.	Skim latex → De ammoniation → Addition of dilute sulphuric acid (20-50%) →coagulation → washing →drying → packing →despatching. Used for the manufacture of low quality rubber articles.	3	3	3
3.	Sheets dried in hot air are called ADS. The chamber is heated by blowing air over a set of electric heaters to which steam is supplied from a boiler. Drying temp for the first two days is limited to 37-40°C otherwise sheets may have a darker shade. During the last two days temperature raised to 72-75°C	3	3	3
4.	Collection → sieving/bulking → standardisation of latex → addition of chemicals → coagulation → sheeting →dripping → drying	3	3	3
5.	No.1X RSS must be produced under conditions where all processes are carefully and uniformly controlled. The rubber must be dry, clean, evenly smoked, free from blisters , sand and any other foreign matter	3	3	3
6.	Pale Latex crepe is the crepe rubber which is produced	3	3	3

	from latex.Grades: No.IX thick/thin, No.1 thick/thin, , No.2 thick/thin, , No.3 thick/thin.			
7.	Oil Extended Natural Rubber: If the concentration of oil added is in the range of 20-40 phr the oil function as an extender for cheapening the product. it can be done in two ways a) in the latex stage- by adding an aqueous emulsion of oil in to latex and coagulating the mixture using an acid b) In the dry rubber stage- naphthenic or aromatic oil and rubber are mixed in a specially modified extruder.[any one method]	3	3	3
8.	Latex as dry rubber(30%), unsmoked sheet (30%),field coagulam (30%),plasticizer (10%).	3	3	3
9.	The reclamator machine consist of a extrusion screw with a hopper at one end into which previously oil and chemical mixed crumb is fed. It generates its own heat for de polymerization by mechanically working the finely devided rubber crumb under pressure and then discharged as reclaimed rubber.	3	3	3
10.	SP rubber production: Preparation of dispersion of vulcanising agent → vulcanization of latex → blending of vulcanized latex with ammonia preserved field latex →coagulation of the blend → processing of the coagulam.	3	3	3
III	PART C			
1.	Methods of application of Yield stimulants: Bark Application-Stimulant is applied on a strip of scrapped bark below the tapping cut. Panel Application-Stimulant is applied on the renewing bark just above the entire length of the tapping cut. Groove Application-Tree lace is removed and stimulant is applied on the tapping cut after wiping away any latex that is oozed out. Lace Application-Stimulant is applied on the tapping cut without removing the lace.	7	7	7

	Soil Application- Stimulant is directly buried in the soil near the tree base, or placed on the soil surface			
	OR			
2.	Intensive tapping: This can be done by increasing the number and length of tapping cut and by increasing the frequency of tapping to extract maximum latex Slaughter Tapping: last stage implemented in the tapping cycle implemented two three years before replanting to extract maximum latex. Ladder tapping: Tapper uses a small ladder to reach the high panel which is opened when tapping on base panel is uneconomic. Puncture Tapping: Extracting latex through punctures made on stimulated bark. Controlled Upward tapping: high panel exploitation using a modified gauge knife.	7	7	7
3.	The different steps involved in polythene skirt-guarding are as follows; 1. Frill the polythene at equal distance using a sewing machine, ensuring 40 percent reduction in length. 2. Using a scraper, lightly scrape the dry bark 10 cm above the tapping cut and parallel to it. The scraped band should extend a minimum of 15 cm from front and back end of the tapping cut. 3. After removing the dust, firmly smear a thin film (roughly 2 mm thick and 4 cm wide) of rain-guard compound on the scraped band. 4. Fix the frilled polythene on this, ensuring that only the lower half (2 cm) of the rain-guard compound is covered by the polythene. 5. Fix the 'kora' cloth ribbon of around 2 cm width firmly using suitable stapler pins over the cut end of the polythene in such a way that 1 cm width of the cloth will be over the polythene and the remaining portion over the compound. 7. Then apply the second coat of the compound to completely cover the cloth ribbon.	7	7	7
	OR			

4.	A group of plants produced by vegetative propagation from a single tree is termed as a clone. Based on origin : Clones classified as 1) primary- clones from mother trees of unknown parentage. Eg: RRII-5, PR-107 2)Hybrid..Two types a) Secondary- they are developed by crossing two primary clones Eg: RRII- 105, RRII 203 etc. b) Tertiary- They are produced by controlled pollination two clones which are either from secondary clones or at least one from secondary clones. Eg: PV 311, PV 235.				7	7	7
5.	The plant shall be located in a place which permits gravity flow of latex from one stage to the next. The factory floor should be in different levels. Field latex storage tanks must be placed at the top most floor. The centrifuging machines are placed on the floor just below it. Cenex tanks are placed on the floor next to machine floor. Skim collection tanks are situated a little away from the main factory floor to minimise bacterial contamination.				7	7	7
	OR						
6.	Characterstics	HA	MA	LA	7	7	7
	DRC (MIN)	70	70	70			
	NRC(max)	2	2	2			
	COAGULAM CONTENT(max)	0.05	0.05	0.05			
	SLUDGE CONTENT(max)	0.1	0.1	0.1			
	ALKALINITY (MIN)	0.7	0.3-0.7	0.3			
	KOH NO. (max)	1	1	1			
	MST(MIN)	475	475	475			
	VFA(max)	0.15	0.15	0.15			
	COPPER CONTENT(max)	8	8	8			
	MANGANESE CONTENT(max)	8	8	8			
7.	Latex is collected and ammoniated to 1.0 to 1.2 per cent by weight and kept in a few weeks for ageing. The creaming agent solution (3%) and soap(10%)are prepared and sieved. The preserved and de-sludged latex is taken in the tank. The required quantity of soap solution and then the creaming agent solution is added to the latex slowly with stirring. After complete addition of creaming agent						

	solution, stirring has to be continued for about 1 h. The creaming tank is closed and the latex is allowed to remain undisturbed till the desired level of creaming is obtained. Although a minimum period of 48 h is usually required for satisfactory creaming no fixed time can be set for all conditions. When the desired level of separation is obtained, the skim layer is drained off through the outlet valve at the bottom. The concentrate obtained is homogenized by stirring, the DRC and ammonia content is adjusted and the latex packed.	7	7	7
	OR			
8.	Production: Centrifuged latex (30% DRC) —————→ Centrifuging (0.8% NRC reduction) —————→ Standardize —————→ Preservation Storage & Despatch (Give description) Application: Production of products like condoms, surgical gloves etc which require a purer form of latex.	6 		

	latex is treated with hydroxylamine neutral sulphate. Latex is then coagulated and processed.			
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Mark Distribution

Module	Hr/ Module	Marks/ Module ($h_i / \Sigma H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
1	11	31.5	2	2	1	3	4	28	7	33
2	12	34.3	3	3	1	3	4	28	8	34
3	10	28.6	2	2	4	12	2	14	8	28
4	10	28.6	2	2	4	12	2	14	8	28
Total	43 (+2hours for internal exam)	123.0	9	9	10	30	12	84	31	123

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COGNITIVE LEVEL MARK DISTRIBUTION[Cognitive wise question analysis]

Cognitive Level	Marks	% of Marks
Remembering	42	34
Understanding	81	66
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

QUESTION WISE ANALYSIS

Course:

Qn. No.	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.04	Remembering	1	2
I.2	M1.02	Remembering	1	2
I.3	M2.03	Remembering	1	2
I.4	M2.04	Remembering	1	2
I.5	M3.02	Remembering	1	2
I.6	M3.04	Remembering	1	2
I.7	M3.04	Remembering	1	2
I.8	M4.02	Remembering	1	2
I.9	M4.04	Remembering	1	2
II.1	M1.01	Remembering	3	7.5
II.2	M2.04	Understanding	3	7.5
II.3	M3.02	Understanding	3	7.5
II.4	M3.02	Understanding	3	7.5
II.5	M3.02	Remembering	3	7.5
II.6	M3.03	Remembering	3	7.5
II.7	M4.04	Understanding	3	7.5
II.8	M4.04	Remembering	3	7.5
II.9	M4.05	Understanding	3	7.5
II.10	M4.02	Understanding	3	7.5
III.1	M1.04	Remembering	7	17
III.2	M1.03	Understanding	7	17
III.3	M1.04	Understanding	7	17
III.4	M1.02	Understanding	7	17
III.5	M2.03	Understanding	7	17
III.6	M2.03	Remembering	7	17
III.7	M2.03	Understanding	7	17
III.8	M2.04	Understanding	7	17
III.9	M3.04	Understanding	7	17
III.10	M3.04	Remembering	7	17
III.11	M4.03	Understanding	7	17
III.12	M4.03	Understanding	7	17

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

NATURAL RUBBER PRODUCTION
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List the major types of rain guards popular in India?	M 1.04	R
2	Define rubber clones?	M 1.02	R
3	What is skim latex?	M 2.04	R
4	What are the ingredients of LATZ preservative system?	M 2.02	R
5	What are the different types of crepe rubber?	M 3.03	R
6	What are the dry marketable forms of Natural Rubber?	M 3.01	R
7	Name the different grades of Ribbed Smoked Sheets?	M 3.02	R
8	Name any two speciality natural rubbers?	M 4.01	R
9	Write any two properties of Oil Extended Natural Rubber?	M 4.04	R

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Mention the nature of action of yield stimulants?	M1.04	U
2	List out the reasons for concentration of NR latex?	M2.03	U
3	Enlist the advantages of ISNR?	M3.04	U

4	Give a short note on the manufacturing process of Air Died Sheet?	M3.02	U
5	Mention the main steps in the production of block rubber?	M3.04	U
6	List the BIS specifications for ISNR 5 ?	M3.04	R
7	Give the main steps in the production process of Constant Viscosity rubber?	M4.03	U
8	What are the properties of tyre rubber?	M4.04	U
9	Write the main steps in the preparation of Low Viscosity Rubber?	M4.03	U
10	What is SP rubber? Give an example?	M4.02	R

PART C

VI. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain a suitable method to measure standard of tappability of rubber trees?	M 1.03	U
OR			
2	Differentiate tapping rest and tapping task?	M 1.03	R
3	Mention the various propagation methods adopted for rubber plants?	M 1.02	U
OR			
4	Explain different methods of applications of yield stimulants?	M 1.04	R
5	Comment on the composition and colloidal nature of NR latex?	M2.01	U
OR			
6	Illustrate the centrifuging process with a diagram of centrifuging machine?	M2.03	U
7	Explain various short term and long term preservative systems for NR latex?	M2.02	R
OR			

8	Explain the role of microorganisms in destabilization and pre coagulation of latex?	M2.02	U
9	Describe the manufacture of RSS with a flow diagram and enlist different grades of it?	M3.02	U
OR			
10	Explain the production of Estate Brown Crepe?	M3.03	U
11	Summarize the production process and important properties of PA 80 rubber?	M4.02	U
OR			
12	List the advantages and applications of reclaimed rubber?	M4.05	R

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SCHEME OF VALUATION

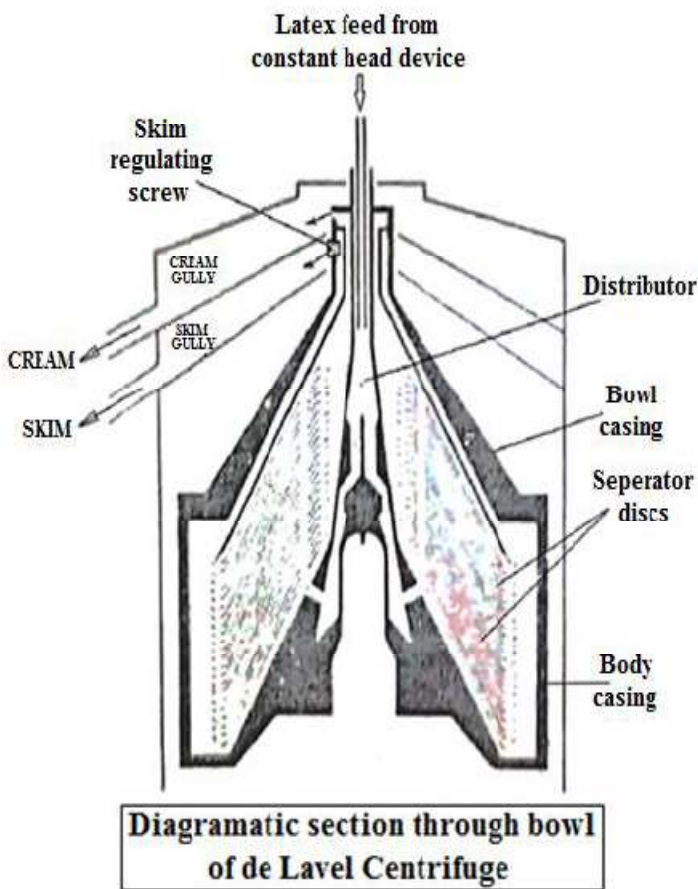
(SCORING INDICATORS)

Revision : 2021					Course Code :				
Course Title : Natural rubber production									
Qn. No.	SCORING INDICATORS				SPLIT UP SCORE		SUB TOTAL		TOTAL
I	PART A								
1.	Polythene Skirt, Modified ready to use Polythene Skirt, tapping shade etc[any two]				0.5*2		1		1

2.	A group of plants produced by vegetative propagation from a single tree is termed as a clone.	1	1	1
3.	When field latex is centrifuged a large volume of serum containing a small portion of rubber content is obtained(2.5-10%) This is known as skim latex	1	1	1
4.	Ammonia, TMTD, ZnO and Lauric Acid	0.25*4	1	1
5.	Pale latex crepe and estate brown crepe	0.5*2	1	1
6.	Block rubber, crepe rubber, crumb rubber	0.5*2	1	1
7.	Grades: RSS IX, 1,2,3,4,5	1	1	1
8.	Oil extended Natural rubber, Tyre rubber	0.5*2	1	1
9.	- Increase in oil content cause high skid resistance on wet surfaces when used in tyre tread compounds - Better grip on snow and ice	0.5*2	1	1
II	PART B			
1.	Yield stimulants produce ethylene and stimulation delays latex vessel plugging and prolongs duration of latex flow.	3	3	3
2.	Transport economy, Standardization of latex as an industrial raw material, purification of latex by removing water soluble NRC, industry requires high DRC for latex product manufacturing etc.	3	3	3
3.	Quality assurance with respect to important technical parameters, consistency in quality, clean ,easy to handle, less storage space, impurities in the rubber can be assessed, less chance of contamination, consumer able to choose the grade according to requirement, minimum variation of properties within same grade. etc.(any six)	0.5*6	3	3
4.	Sheets dried in hot air are called ADS. The chamber is heated by blowing air over a set of electric heaters to which steam is supplied from a boiler. Drying temp for the first two days is limited to 37-40°C otherwise sheets may have a darker shade. During the last two days temperature raised to 62-65°C.	3	3	3
5.	Latex → Reception, Sieving&Bulking → Coagulation→ Coagulam Crusher/Prebreaker → Crepers(2 to 4Nos.) → Hammer Mill/Shredder→ blending→ Drying →	3	3	3

	Pressing→ Testing → Grading→ Packing																	
6.	<table><tr><th>Characterestics</th><th>ISNR 5</th></tr><tr><td>Dirt Content</td><td>0.05</td></tr><tr><td>Volatile matter</td><td>0.8</td></tr><tr><td>Ash content</td><td>0.6</td></tr><tr><td>Nitrogen Content</td><td>0.6</td></tr><tr><td>Initial Plasticity(P₀)</td><td>30</td></tr><tr><td>PRI</td><td>60</td></tr></table>	Characterestics	ISNR 5	Dirt Content	0.05	Volatile matter	0.8	Ash content	0.6	Nitrogen Content	0.6	Initial Plasticity(P ₀)	30	PRI	60	3	3	3
Characterestics	ISNR 5																	
Dirt Content	0.05																	
Volatile matter	0.8																	
Ash content	0.6																	
Nitrogen Content	0.6																	
Initial Plasticity(P ₀)	30																	
PRI	60																	
7.	CV Rubber Production: Blending of NR → Bulking → Diluting → Ammoniation → Addition of 0.15 phr hydroxylamine hydrochloride → Coagulation → Processing into marketable form	3	3	3														
8.	Properties: 1. Resistance to low temperature crystallization 2. low mooney viscosity 3. good mechanical properties 4. good flex cracking resistance 5. low heat build up	3	3	3														
9.	LV rubber: Naphthanic oil (4phr) on DRC of latex is added to the latex in the form of emulsion. Before that latex is treated with hydroxylamine neutral sulphate. Latex is then coagulated and processed	3	3	3														
10.	Intimate mixture of vulcanized latex with field latex in different proportions before coagulation is Superior Processing rubber(SP) Eg.PA 80	3	3	3														
III	PART C																	
1.	Rubber trees are tapped only after they attain a standard girth. Its measured using a standard girth wire of 50 or 55cm long attached to a straight stick. Seedling plants- 50 cm height and 55cm girth or 90cm height and 50cm girth from ground level. Budded Trees- 125cm height and 50cm girth from the b union	7	7	7														

	OR			
2.	Tapping Rest: Rubber trees shed their leaves during December to February and refoliate soon after wintering. During this time yield will be poor and trees are given about four weeks rest. Tapping task: Number of trees assigned to a tapper for a day's tapping. Its based on the stand per ha and topography of land. Normally in India, it varies from 300-400. The number reduces depending on double cut or intensive systems are practiced.	3.5*2	7	7
3.	Poly clonal seed germination: Several clones are planted intermixed in a garden to maximise cross pollination. Vegetative methods; Budding- buds from high yielding mother plants grafted to seedling stocks. Brown budding- budding using bud wood of one year old. Green budding- buds from 6-8 weeks old bud wood which are green in colour. other methods: tissue culture, layering, stem cutting etc.	7	7	7
	OR			
4.	Methods of Application o Yield stimulants: Bark Application- Stimulant is applied on a strip of scrapped bark below the tapping cut. Panel Application- Stimulant is applied on the renewing bark just above the entire length of the tapping cut. Groove Application- Tree lace is removed and stimulant is applied on the tapping cut after wiping away any latex that is oozed out. Lace Application- Stimulant is applied on the tapping cut without removing the lace. Soil Application- Stimulant is directly buried in the soil near the tree base, or placed on the soil surface.	7	7	7

5.	Composition of NR latex: Rubber - 30 - 40%, Protein – 1 - 1.5%, Resin- 1 - 2.5%, Sugar - 1%, Ash - <1%, Water- 55 - 60% The proteins and phospho lipids are responsible for the hydro philicity by forming a thin layer over its surface. It helps in acquisition of - ve charge to the particle which increase the stability of the latex.	3.5	7	7
	OR			
6.	 Diagrammatic section through bowl of de Laval Centrifuge Latex feed from constant head device Skim regulating screw CREAM GULLY SKIM GULLY CREAM SKIM Distributor Bowl casing Seperator discs Body casing Latex is subjected to a strong centrifugal force several thousand times of the gravitational force in a bowl rotating at a high speed where by individual rubber particles tend to separate in to a layer surrounding the axis of rotation. The skim flows outwards away from the axis of rotation.	5	7	7
		2		

7.	When latex is to be stored for a short period of time(few hrs or days) short term preservatives are used. Eg: Ammonia (0.01-0.05% as gas or 25% solution) Sodium sulphite (0.05% as 5% solution) Formalin(0.02% as 2% solution) If latex is to be stored for a long period of time long term preservation system is used. Eg Ammonia (0.7-1.0% as gas or 25% solution) LATZ system(0.2% Ammonia, 0.0125% TMTD, 0.0125% ZnO, 0.05% Lauric Acid)	7	7	7
	OR			
8.	Latex collected under field conditions get contaminated by different bacteria and yeast. Some bacteria feed on the carbo hydrates or sugar and convert them in to volatile fatty acid such as formic acid and acetic acid. Some other bacteria attack the proteins in latex and decompose them to simple products such as H₂S and SO₂, causing bad odour. This process is termed as putrefaction. The VFA and these components leads to progressive destabilization of latex, it results in premature coagulation or spontaneous coagulation of latex.	7	7	7
9.	Collection → sieving/bulking → standardisation of latex → addition of chemicals → coagulation → sheeting →dripping → drying.(Give Description) Grades: RSS IX, 1,2,3,4,5	6 1	7	7
	OR			
10.	Estate brown crepe: Collection of field coagulam→ Soaking → Filtering → Crepe making with Washing (Through Creping Battery/.Mascerators) → Drying → Weighing & Baling. (Give Description)	7	7	7

11.	PA 80: - Consists of 80 parts of vulcanized and 20 parts of unvulcanized latex. - Vulcanized latex is first diluted to 20% of TSC. - This is then mixed with fresh field latex (1:4) - Sieved - Coagulate with 5% sulphuric acid - Coagulum soaked in water for 30 min - Crepe making - Drying at 70 °C - Pressing into bales Properties: - Good extrusion properties - Low die swell - Higher stock viscosity, easy to handle	5.5	7	7
	OR			
12.	Advantages: - It is economic. Reclaim rubber added compounds are much cheaper. Even in a high quality compound, reclaimed rubber if used in judicious proportions can give considerable savings. - Ease of processibility in the mixing mill, extruder, calender, frictioning, and skim coating operations. - Reclaimed rubber is uniform and is compatible with most accelerators - Faster curing compounds can be made with reclaimed rubber. Applications: Reclaim rubber is primarily used in producing mechanical rubber goods. It is also used in tyres, inner tubes, tyre lining, tyre repair, retreading, general moulding, belting, adhesives, mastics, footwear, sheeting, matting, belting, cable bedding compound, and sound reduction.	1*4	7	7
		3		

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Mark Distribution

Module	Hr/ Module	Marks/ Module ($h_i / \Sigma H_i$) *123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
1	11	31.5	2	2	1	3	4	28	7	33
2	12	34.3	3	3	1	3	4	28	8	34
3	10	28.6	2	2	4	12	2	14	8	28
4	10	28.6	2	2	4	12	2	14	8	28
Total	43 (+2hours for internal exam)	123.0	9	9	10	30	12	84	31	123

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COGNITIVE LEVEL MARK DISTRIBUTION[Cognitive wise question analysis]

Cognitive Level	Marks	% of Marks
Remembering	43	35
Understanding	80	65
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

QUESTION WISE ANALYSIS

Course:

Qn. No.	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.04	Remembering	1	2
I.2	M1.02	Remembering	1	2
I.3	M2.04	Remembering	1	2
I.4	M2.02	Remembering	1	2
I.5	M3.03	Remembering	1	2
I.6	M3.01	Remembering	1	2
I.7	M3.02	Remembering	1	2
I.8	M4.01	Remembering	1	2
I.9	M4.04	Remembering	1	2
II.1	M1.04	Understanding	3	7.5
II.2	M2.03	Understanding	3	7.5
II.3	M3.04	Understanding	3	7.5
II.4	M3.02	Understanding	3	7.5
II.5	M3.04	Understanding	3	7.5
II.6	M3.04	Remembering	3	7.5
II.7	M4.03	Understanding	3	7.5
II.8	M4.04	Understanding	3	7.5
II.9	M4.03	Understanding	3	7.5
II.10	M4.02	Remembering	3	7.5
III.1	M1.03	Understanding	7	17
III.2	M1.03	Remembering	7	17
III.3	M1.02	Understanding	7	17
III.4	M1.04	Remembering	7	17
III.5	M2.01	Understanding	7	17
III.6	M2.03	Understanding	7	17
III.7	M2.02	Remembering	7	17
III.8	M2.02	Understanding	7	17
III.9	M3.02	Understanding	7	17
III.10	M3.03	Understanding	7	17
III.11	M4.02	Understanding	7	17
III.12	M4.05	Remembering	7	17